Biospectra : ISSN: 0973-7057, Vol. 19(2), September, 2024, pp. 215-220 **Plant Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Floristic diversity study of ornamental plant and their parts use in disease around Chapra (Saran) Bihar

Md Sahid Hussain* & Amrendra Kumar Jha
Department of Botany, Jai Prakash Vishwavidyalaya, Chapra (Saran), Bihar, India

Received : 28th June, 2024 ; Revised : 26th July, 2024
DOI:-https://doi.org/10.5281/zenodo.15112002

Abstract- Ornamental plants have been cultivated since ancient times for decorative purposes in garden and landscape design, as house plants cut flowers and patterns and are associated with people. The ornamental plant market is growing but is still affected by the changes caused by climate change in Chapra (Saran). Therefore, diversification of ornamental plants is an important source of breeding material for development of new varieties. To respond to changes in garden design and market search for different types of flowers. Ganga region and Gandak are also areas where ornamental plants are cultivated near Nayagaon town Dighwara and Chapra. There are also many different ornamental plant species and their conservation in national parks, botanical gardens and arboretums by *in situ* or *ex situ* methods.

Key words: Biodiversity, Conservation, Exploration, Ornamental Plants, Cut flower, Diversification, Aesthetic feature, Landscape, Decorative, Cultivated, Annual, Biennial, Perennial

INTRODUCTION

The study area is rich in a wide variety of ornamental plants.[1] These plants are planted in gardens or public areas for beautification and decoration purposes such as ornamental cut flowers and indoor plants.[2,3] These types of plants are roses, lilies, ornamental plants. They are divided into two groups as herbs, vines and shrubs and can be annual and perennial, which adds a lot of color to any garden and can be used as cut flowers in vases. Cutting flowers from plants can provide us with economic benefits as they are used in various flowers.[1] Conservation of the biodiversity of ornamental plants in the world is a process that has developed for human benefit in response to changes in garden design and commercial research on different flowering plants. Ornamental plant diversity is an important breeding material for breeding new varieties.[4]

*Corresponding author :
Phone : 9608682840
E-mail : mdsahid786@gmail.com

Ornamental plants respond to local plants and are known and illustrate the planting and eating ideas of many people in different cultural contexts. The beautiful patterns, colors and textures of stems, leaves, flowers and fruits of ornamental plants are responsible for maintaining the natural beauty and attractiveness of the landscape.[5] Also, introduction and cultivation of beautiful plants and herbs in private and public areas supports the current approach to conservation of endogenous plants in these areas.[3] They are used for decoration and landscaping purposes such as cut flowers and patterns to beautify indoor plants.[3] Ornamental plants provide many ecosystem services important for human health. They also play an important role in supporting and maintaining biodiversity and also enhance the beauty of indoor and outdoor spaces.

MATERIALS & METHODS

This study was carried out at Chapra Saran Bihar India, through July 2022-2023. Using sites investigation in combination with analysis of relevant literature and available data, the present study will include the botanical exploration and investigation of cultivated and wild ornamental plants of the Saran district. There is an abundance of wild and cultivated ornamental plants in district Saran and exhibits a wide range of diversity in terms of taxa, habit, and growth forms.

Floristic diversity study

Work in the period 2022-2023. Taxonomic identification, data and characteristics of each species of ornamental plants were recorded in Floral Art. The method used is based on the analysis of flora identification. All collections were made with the help of current data on local flowers collected by Hasan and Rahma (2021)[6], Gamble and Fisher (1956)[7], Henry *et al.* (1989)[8] and Mathew (1999)[9]. Haynes Flora was also consulted to identify the plants recorded in the different study areas. There are 46 species of ornamental plants in the study area.

Result and Discussion

(Table 1) Among them, there are 10 species of annual plants (5.1%). There are 8 species of biennial plants (4.8%) and 35 species of perennial plants (54%). and bioaesthetic planning are the latest trends in creating environmentally friendly residents. Here we collect plants for ornamental purposes. We would like to say that the study, collection and protection of ornamental species are also a way to control different species and prevent rare and endangered diseases of ornamental value. In recent years, ornamental plants have gained importance in the Indian export market for various uses for the income of the poor and for the income of the poor. Ornamental species are also a source of medicinal value.[10] Ornamental plants in the broadest sense include plants grown for ornamental purposes in gardens, buildings, landscape design projects, squares, parks, street trees, indoor plants and cut flowers. Aesthetic goals respond to the aesthetic values attributed to them by people in different cultures. For example, in Western culture, it is often associated with certain plants: flowers, fruits, leaves, petals, color and fragrance.[3,11]

Table 1- Ornamental plants parts Use in Disease Treatment

Sl. No.	Scientific Name	Family	Annual/ Biennial/ Perennial	Parts Use	Use in Disease
1.	*Ruellia tuberosa*	Acanthaceae	Biennial	Leaves, Roots	Gonorrhea, Syphilis, Bronchitis, Renal diseases
2.	*Capparis zeylanica*	Capparaceae	Perennial	Leaves	Piles, Antodote to snake
3.	*Combretum indicum*	Combretaceae	Perennial	Whole parts	Ridding the body of parasites, Ulcer, fever.
4.	*Senna sophera*	Fabaceae	Annual	Leaves	Antiasthamatic Anti-inflammatory Antioxidant Anti- diabetic
5.	*Solanum virginianum*	Solanaceae	Perennial	Stem, Flower, Fruits.	Epilepsy, Headache, Migraine.
6.	*Argemone mexicana*	Papaveraceae	Biennial	Seeds	Tumors Warts Rheumatism Leprosy, Malaria
7.	*Zephyranthes citrina* Baker	Amaryllidaceae	Perennial	Leaves, Flower, Stem	Diabetes mellitus Tumours
8.	*Dieffenbachia sequine*	Araceae	Perennial	Leaves, Root	Gargle for angina, gout, Vaginal itching
9.	*Asystasia gangetica*	Acanthaceae	Perennial	Leaves	Anthelmintic. Intestinal worms
10.	*Ixora finlaysoniana* Wall	Rubiaceae	Perennial	Leaves, Flower, Bark	Skin diseases Anticancer, Antiasthamatic, Antiviral. Antimicrobial
11.	*Nerium olender splendens*	Apocynaceae	Perennial	Plants juice, Oil	Cardiac illness, Asthma, Diabetes, Cancer.
12.	*Loenurum japonicas* Houtt.	Laminaceae	Annual/Biennial	Stem, Leaves	Dysmenorrhea, Amenorrhea, Menstrual & Delivery disorders.
13.	*Morning glory*	Convolvulaceae	Annual	Seeds	High BP, Diabetes, Heart disease
14.	*Jasminum multiflorum*	Oleaceae	Perennial	Dried leaves, Roots	Antidote for Snake bite, ulcer
15.	*Rauwolfia serpentine*	Apocynaceae	Perennial	Roots	Insomnia, BP, Fever
16.	*Datura stramonium*	Solanaceae	Annual /Perennial	Fruits, Scalp	Inflammation, Stomach treatment
17.	*Gloriosa superba*	Colchicaceae	Perennial	Leaves	Head lice, Reduce labour pain
18.	*Heuchera sanguinea*	Saxifragaceae	Perennial	Root	Heart disease
19.	*Cycas circinalis*	Cycadaceae	Perennial	Leaves, Root, Seeds, Stems	Intestinal disorders, Blood vomiting
20.	*Berberis vulgaris*	Berberidaceae	Perennial	Leaves, Stem	Kidney Stone, Liver disorders
21.	*Mammillaria beneckei*	Cactaceae	Perennial	Stem, Scales	Immune boosters, increase for WBC
22.	*Cissus quadrangularis*	Vitaceae	Perennial	Stem, leaves	Fracture healing in bone, Pain, Swelling
23.	*Achilea millefolium* L.	Asteraceae	Perennial	Leaves	Viral-covid-19,
24.	*Bridelia micrantha*	Phyllanthaceae	Perennial	Bark, Leaves, Root	Wound, cough, Sore throat
25.	*Gymnanthemum amygdalium*	Asteraceae	Perennial	Root, Bark, Flower	STDs, Infertility, Malaria, Diabetes, GIP
26.	*Chrysanthemum indicum*	Asteraceae	Perennial	Flower	Anti-cancer, anti- microbial,
27.	*Nyctanthes arbor-tristis*	Oleaceae	Perennial	Leaves	Arthritis, Fever.
28.	*Rosa hybrid*	Rosaceae	Perennial	Flower	Inflammatory
29.	*Hibiscus rosa sinensis*	Malvaceae	Perennial	Flower	Heart and Nerve diseases
30.	*Gardenia jasminoides*	Rubiaceae	Perennial	Flower	Diabetes, BP, Anxiety
31.	*Hylotelephium spectabile*	Crassulaceae	Perennial	Leaves	High Blood pressure, wound healing
32.	*Hibiscus moscheutos*	Malvaceae	Perennial	Root and Leaves	Gonorrhea, foot disease

33.	*Armeria maritime*	Plumbaginaceae	Perennial	Flower	Heart, Brain, Spinal cord
34.	*Helianthus maximiliani*	Asteraceae	Perennial	Root and Leaves	Autoimmune Diseases
35.	*Angelica archangelica*	Apiaceae	Biennial	Leaves, Seeds and Shoot	Cancer & Blood Disorder
36.	*Campanula rapunculoides*	Campanulaceae	Biennial	Stamen	Upper respiratory disorder
37.	*Heuchera americana*	Saxifragaceae	Perennial	Floral parts	Black Band disease
38.	*Oenothera biennis*	Onagraceae	Biennial	Stem, Root and leaves	
39.	*Gaillardia pulchella*	Asteraceae	Biennial	Root	Venereal diseases, for kidney
40.	*Satureja hortensis* L	Lamiaceae	Biennial	Leaves and Stem	Diarrhea, stomach pain, Cough
41.	*Echinacea purpurea*	Asteraceae	Annual	Leaves and Seed	Common cold, Bronchitis, Coughs.
42.	*Geranium dissectum*	Geraniaceae	Annual	Root	Diarrhea, Anti- diarrhea,
43.	*Beginia oblique*	Begoniaceae	Annual	Leaves	Skin diseases.
44.	*Eschscholzia sp.*	Papaveraceae	Annual	Aerial parts	Encephalitis
45.	*Bellis perennis*	Asteraceae	Annual	Aerial parts	Bleeding, Bronchitis
46.	*Aster amellus*	Asteraceae	Annual	Whole parts	Pain, Fevers, Diarrhea.




Ixora finlaysoniana
Nerium olender splandens
Argemone maxicana
Asystasia gangetica
Solanum verginianum
Capparis zeylanica
Senna sophera
Cobretum indicum
Ruellia tuberosa
Datura stramonium
Morning glory
Delonic regia
Rauwolfia serpentine
Loenurum japonicas Houtt.

CONCLUSION

Finally, it may be concluded that such study of floristic diversity study of ornamental plants will support for use of plant parts for disease qureable, conservation and protection as well as large scle production of industrial raw materials for pharmaceuticals. And dominant ornamental plants for income of poor family. Study further suggests that rare abundant ornaental plants should be conserved them and protect in their natural envuroment. Rare ornamental plants is required their conservation for future generation for medicinal uses

REFERENCES

1. **Kensa V. M. Ancelsowmiya S., Jijomickle J. Meera S. Sindhu J. Sumathi M. S., Radhika R., 2014.** Exploration of ornamental floras in the campus of bishop's house, ngercoil, Kanyakumari district, India. *Int. J. Curr. Microbiol App. Sci.* **3(10):** 441-448.
2. **Kumar M., Sharma G., and Alka D., 2014.** Medico ethno botany of plants surveyed and studied in district bijnor with special emphasis on their medicinal, religious and ornamental significance. *Indian J Appl. Res*. **4(6):** 40-43.
3. **Oloyede F. A., 2012.** Survey of ornamental ferns, their morphology and uses for environmental protection, improvement and management, *Ife J. Sci.,* **14(2):** 245-252
4. **Chowdhuri K.T, Deka K. 2019**. Biodiversity and conservation of ornamental crop. PP 139-21.
5. **Reddy S. R., Reddy A. M. and Yasodamma N., 2012.** Exploration of wild ornamental flora of Ysr district, Andhra Pradesh India. Indian *J. Fund. App. Life Sci*., **2(1):** 192-199.
6. **Islam. Md., T. Hasan. Md. M., Rahman. Md. A., Das. S. K. and Sultana T., 2021.** Floral diversity of abandoned mansions and the influence of soil properties on these unique vegetation. *Aemrican J.P Plant Sci*. **12(10):**1559-1600.
7. **Gamble J. S. and Fischer C. E. C., 1956.** Flora of the presidency of Madras.vol. 1-3 Adlard and Sons Ltd London.
8. **Henry A. N., Chithra V., and Balakrishnan N. P., 1989.** Flora of Tamilnadu. Series-1.vol. 3, Botanical Survey of India, Coimbatore.
9. **Mathew K. M., 1999.** The Flora of the palani hills South India. Vol. 3 the Rapinat Herbarium, Thiruchirapalli. p1195.
10. **Aasati B. S. and Yadav D. S., 2010.** Diversity of horticultural crops in North Eastern region. *ENVIS Bulletin: Himalayan Ecol,* **12(1):** 1-11.
11. **Estrada-Castillon E., Garza-Lopez M., Villarreal-Quintanilla J. A., Salinas-Rodriguez. M. M., Soto-Mata B. E., Gonzalez-Rodriguez H., Gonzalez-Uribe D. U., Cantu-Silva I., Carrillo- Par A., Cantu-Ayala C., 2014.** Ethanobotany in rayones nuevo Leon Mexico. *J. Ethanobiolo. Ethanomedicine*, **10:**62.

***